Learning a ~~function on a~~ manifold

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1 Definitions and Previous Results

Definition 1. *A metric tensor on (the embedded in $\mathbb{R}^k$) manifold $\mathcal{M}$ is a function $g : \mathcal{M} \to \mathbb{R}^{k\times k}, \forall x \in \mathcal{M}, \operatorname{rank} g(x) = \dim \mathcal{M}, g(x) \succeq 0$.*

Definition 2 (Dirichlet energy)**.** *The* Dirichlet energy *of a function $f : \mathcal{M} \to \mathbb{R}$ with respect to the manifold $\mathcal{M}$ equipped with the metric tensor g is defined as*

$$\mathcal{E}_{\mathcal{M},g}(f) = \frac{1}{2}\int_{\mathcal{M}} \|\nabla_g f(x)\|_g^2 \, d\mu_g(x),$$

where $\nabla_g f$ denotes the gradient of f with respect to the metric g, $\|\cdot\|_g$ denotes the norm induced by g, and μ_g is the volume measure associated to g on $\mathcal{M}$.

Definition 3 (Measure)**.** *A Riemannian metric on a d-dimensional manifold is a smoothly varying inner product*

$$g_x : T_x M \times T_x M \to \mathbb{R}.$$

In local coordinates, it is represented by a positive-definite matrix $g(x) = [g_{ij}(x)]$. The metric induces the Riemannian volume measure, often denoted:

$$d\mu_g(x) = \sqrt{\det g(x)} dx^1 \cdots dx^d.$$

This is the analogue of "area $= \sqrt{\det g}$" in differential geometry. $d\mu_g(x) = \sqrt{\det g(x)} dx^1 \cdots dx^d$.

Definition 4 (Gradient)**.** *The gradient of a function $f : \mathcal{M} \to \mathbb{R}$ with respect to the metric tensor g is defined as*

$$\nabla_g f(x) = g(x)\nabla f(x) \in T_x\mathcal{M},$$

where $\nabla f(x)$ is the gradient of f with respect to the Euclidean metric.

Definition 5 (Laplacian)**.** *The Laplacian of a function $f : \mathcal{M} \to \mathbb{R}$ with respect to the metric g is defined as*

$$\Delta_g f(x) = \nabla \cdot \nabla_g f(x) = \nabla \cdot (g\nabla f(x)) \in \mathbb{R}$$

Theorem 1 (Strong Whitney embedding theorem)**.** *The* strong Whitney embedding theorem *states that any smooth real m-dimensional manifold (required also to be Hausdorff and second-countable) can be smoothly embedded in the real $2m$-space, $\mathbb{R}^{2m}$, if $m > 0$. This is the best linear bound on the smallest-dimensional Euclidean space that all m-dimensional manifolds embed in, as the real projective spaces of dimension m cannot be embedded into real ($2m - 1$)-space if m is a power of two (as can be seen from a characteristic class argument, also due to Whitney).*

Theorem 2 (weak Whitney embedding theorem)**.** *The* weak Whitney embedding theorem *states that any continuous function from an n-dimensional manifold to an m dimensional manifold may be approximated by a smooth embedding provided $m > 2n$. Whitney similarly proved that such a map could be approximated by an immersion provided $m > 2n - 1$. This last result is sometimes called the Whitney immersion theorem.*

2 Learning a function

Given:

- An embedded manifold $\mathcal{M} \subset \mathbb{R}^k, \dim \mathcal{M} = d$
- A metric tensor on $\mathcal{M}$: $g : \mathcal{M} \to \mathbb{R}^{k\times k}, \forall \mathbf{x} \in \mathcal{M}, \operatorname{rank} g(\mathbf{x}) = d, g(\mathbf{x}) \succeq 0$.
  Note: For simplicity, we assume $\det g(\mathbf{x}) = 1$ hence, the measure is the Lebesgue measure.
- A dataset $D = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where $\mathbf{x}_i \in \mathcal{M}, y_i \in \mathbb{R}$

Goal: to learn the smoothest function $f : \mathcal{M} \to \mathbb{R}$ such that the function $f(\mathbf{x}_i) \approx y_i$.

Constrained optimization:

$$\min_f \mathcal{E}_{\mathcal{M},g}(f)$$
$$\text{s.t. } f(\mathbf{x}_i) = y_i, \forall i \in [n].$$

Unconstrained optimization:

$$\min_f \mathcal{E}_{\mathcal{M},g}(f) + \lambda \sum_{i=1}^{n} (f(\mathbf{x}_i) - y_i)^2 =$$
$$\min_f \frac{1}{2} \int_{\mathcal{M}} \|\nabla_g f(\mathbf{x})\|_g^2 \, d\mu_g(\mathbf{x}) + \lambda \sum_{i=1}^{n} (f(\mathbf{x}_i) - y_i)^2$$

Gradient descent, convex optimization (in a *function* space):

$$\text{PDE: } \frac{\partial f}{\partial t} = -\Delta_g f + 2\lambda \sum_{i=1}^{n} (f(\mathbf{x}_i) - y_i)\delta_{\mathbf{x}_i}$$
$$\text{step: } f_{t+1} = f_t + \eta \left(\Delta_g f_t - 2\lambda \sum_{i=1}^{n} (f(\mathbf{x}_i) - y_i)\delta_{\mathbf{x}_i} \right)$$

Definition 6 (Gradient-flow evolution operator)**.** *We will denote $\Phi_T(f_i)$ as the gradient-flow evolution operator of f_i after time T of gradient descent. Note, that $\Phi_T(f_i) = \Phi_T(f_i, g, \mathcal{M}, D)$ depends on the metric tensor g, the manifold $\mathcal{M}$, and the dataset $D = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$.*

Discretization in space (Disclaimer: this is a bit of a stretch.) If $\mathcal{M}$ is discretized as a point cloud $\mathcal{P} = \{\mathbf{z}_i\}_{i=1}^n$ with weights

$$w_{ij} = \exp\left(-\frac{\|\mathbf{z}_i - \mathbf{z}_j\|_{g(\mathbf{z}_i)}^2}{\epsilon} \right)$$

(epsilon TBD), one may argue that:

$$L_{\mathcal{P}} D_{\mathcal{P}}^{-1} \approx \Delta_g$$

Hence, one step of gradient descent becomes (yes, here comes the "stretch") like a transformer layer:

$$f_{t+1}(\mathbf{z}) = \underbrace{f_t(\mathbf{z})}_{\text{Residual}} + \eta \left(\underbrace{\frac{\sum_i \exp(-\|\mathbf{z} - \mathbf{z}_i\|_{g(\mathbf{z})}^2/\epsilon) f_t(\mathbf{z}_i)}{\sum_j \exp(-\|\mathbf{z} - \mathbf{z}_j\|_{g(\mathbf{z})}^2/\epsilon)}}_{\approx \text{Attention}} - \underbrace{2\lambda \sum_{i=1}^{n} (f(\mathbf{x}_i) - y_i)\delta_{\mathbf{x}_i}(\mathbf{z})}_{\text{Residual}} \right)$$

Idea 1: what if a fixed transformer performs gradient descent on the function space, hence, learns a function in context?

3 Learning a manifold

What if we are given a few smooth functions $f_1, \dots, f_l : \mathbb{R}^k \to \mathbb{R}$ on a manifold $\mathcal{M}$ embedded in $\mathbb{R}^k$. However, we don't know the metric tensor g on $\mathcal{M}$. Can we learn the metric tensor such that the functions $f_1, \dots, f_l$ are as smooth as possible? If we do so, then we can easily learn a new smooth function from a few observations on the fly next time.

For this problem, the objective is the following:

$$\min_g \sum_{i=1}^{l} \mathcal{E}_{\mathcal{M},g}(f_i)$$
$$\text{s.t. } \det g(\mathbf{x}) = 1 \text{ for all } \mathbf{x} \in \mathcal{M}.$$

where $\mathcal{E}_{\mathcal{M},g}(f_i)$ is the Dirichlet energy of f_i with respect to the metric tensor g. If we discard the constraint on the determinant or relax it to $\alpha I \preceq g(\mathbf{x})^{-1} \preceq \beta I$, the problem is convex w.r.t. g.

Okay, as of now, we didn't have a need to perform GD in a function space inside the global optimization with respect to the metric tensor g – there was no need to train a transformer-like model with multiple layers. In what case would that be necessary?

4 Lessons from Kernel Methods

- If we know *the right* kernel $\kappa(x, y)$, given $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ we can approximate any $f(x)$ is one step. For example, in Jacot, Gabriel, and Hongler [JGH20] NN learns a function that would give us the right kernel.
- The generalization error will on
  - the number of samples n
  - the spectrum of $\mathcal{T}_\kappa$
  - the regularity of the f

  as the risk is expressed as

$$\mathcal{E}\left(\widehat{f}_\lambda\right) = \underbrace{\sum_{j\geq 1}\left(\frac{\lambda}{\mu_j+\lambda}\right)^2 f_j^2}_{\text{bias term}} + \underbrace{\frac{\sigma^2}{n}\sum_{j\geq 1}\frac{\mu_j}{(\mu_j+\lambda)^2}}_{\text{variance term}}$$

- So again, given some functions $f_1, \ldots, f_i$ we want to find kernel κ with respect to what those functions are as smooth as possible.
- If we can do that we can easily learn a function on a fly.
- paper Shen et al. [She+25] brigdes kernel methods to $O(1)$-layer transformers.

5 Distorted Manifold Hypothesis

Let μ be a probability distribution supported on a d-dimensional manifold $\mathcal{M} \subset \mathbb{R}^k$.

Hypothesis: True data: set of points $x \in \mathbb{R}^k$ that comes from μ – a probability distribution supported on a d-dimensional manifold $\mathcal{M} \subset \mathbb{R}^k$, $d \ll k$. However, we some distortion process $Z(\mathbf{x}, t)$ we observed distorted points $\widetilde{\mathbf{x}}$ not $\mathbf{x}$. For example, the distortion governed by time-inhomogeneous diffusion $D(t) : [0, 1] \to \mathbb{R}^{n\times n}, D \succeq 0$ and drift $b : [0, 1] \to \mathbb{R}^n$.

$$\begin{cases} \dfrac{\partial \mu_t(x)}{\partial t} = \nabla_x \cdot \big(D(t)\,\nabla_x \mu_t(x)\big) \;-\; b(t)\cdot \nabla_x \mu_t(x), & x \in \mathbb{R}^k,\ t \in (0,1], \\ \mu_{t=0}(x) = \mu(x), & x \in \mathbb{R}^k. \end{cases}$$

For an effective kernel method, we should eliminate some distortion before fitting observed data: $\{(\widetilde{\mathbf{x}}_i, y_i)\}$. I believe that finding *minimal-embedding* of μ helps with that.

Definition 7. *A minimal embedding of μ is the map*

$$\Phi : \mathcal{M} \to \mathbb{R}^k$$

that minimizes the Dirichlet energy of the embedding:

$$\Phi^\star = \arg\min_{\Phi:\mathcal{M}\to\mathbb{R}^k} \mathcal{E}(\Phi) := \frac{1}{2}\sum_{j=1}^k \int_{\mathcal{M}} \|\nabla_{\mathcal{M}} \Phi_j(x)\|^2 \, d\mu(x)$$

subject to natural constraints:

- Φ *preserves the intrinsic dimension (rank of the Jacobian is d almost everywhere),*
- $\mathbb{E}_\mu[\Phi(x)] = 0$ *(remove trivial translations),*
- $\frac{1}{d}\int_{\mathcal{M}} \Phi(x)\Phi(x)^\top d\mu(x) = I_d$

Intuition: the minimal embedding is the least energetic or smoothest way to embed the distribution into $\mathbb{R}^k$. Equivalently:

- it is the embedding whose coordinate functions are harmonic: $\Delta_{\mathcal{M}} f = 0$.
- it removes all geometric distortions caused by drift/diffusion processes,
- it recovers the "canonical" coordinates of the underlying manifold w.r.t. the intrinsic geometry induced by μ.
- Minimal embedding = the embedding with the simplest geometry for which μ looks undistorted.

Idea: Transformer layers reverse the distortion and recoved the minimal energy of the $\widetilde{\mu}$.

For example to reverse the diffusion one should do:

$$\begin{cases} -\partial_t \mu_t(x) = -\nabla\cdot(b_t(x)\mu_t(x)) + \nabla\cdot(D_t(x)\nabla\mu_t(x)), & t \in [0,1], \\ \mu_1(x) = \widetilde{\mu}(x). \end{cases}$$

One transformer layer = One gradient step in the measure space that captures the equation above (on a discretized μ).

However, this process is ill-posed as the diffusion destroys high-frequency models so some information will be inevatably lost.

6 Scaling Laws

This framework can explain the Scaling laws of the transformers as well [Kap+20]:

- Test loss as a power law in model size – how many frequencies can be captured by the model, depends on the spectrum decay rate, how fine discretization of the reversed distortion process is.

$$L(N) = L_\infty + AN^{-\alpha}$$

- Test loss as a power law in dataset size – how many points are sampled from the manifold.

$$L(D) = L_\infty + BD^{-\beta}$$

- Test loss as a power law in compute – how many gradient descent steps are performed on a function $D(t, \cdot)$ in the space of possible distortions.

$$L(C) = L_\infty + KC^{-\gamma}$$

Empirical exponents (Kaplan et al. 2020):

$$\alpha \approx 0.076, \beta \approx 0.095, \gamma \approx 0.048$$

7 The Last Piece of the Puzzle!

Theorem 3 (Takahashi's Theorem (1966) [Tak66]). *Let*

$$X : \left(M^d, g\right) \longrightarrow S^{k-1}(1) \subset \mathbb{R}^k$$

be an isometric immersion of a d-dimensional Riemannian manifold into the unit sphere. Then the following are equivalent:

1. *$X(M)$ is a minimal submanifold of the sphere.*
   That is, its mean curvature vector H (as a submanifold of S^{k-1}) satisfies

$$H \equiv 0.$$

2. *The coordinate functions of X are Laplace-Beltrami eigenfunctions with eigenvalue d :*

$$\Delta_M X = -dX$$

   Here Δ_M acts componentwise on

$$X = \left(X^1, \ldots, X^k\right) : M \to \mathbb{R}^k$$

   Equivalently:

$$\Delta_M X^j = -dX^j, \quad j = 1, \ldots, k$$

So what transformer is doing is

1. Denoising the data via reverse diffusion process.
2. Finding a minimal embedding of the manifold $\mathcal{M}$ into the unit sphere $S^{k-1}(1) \subset \mathbb{R}^k$, hence, its first k eigenfunctions: $\phi_1, \ldots, \phi_k$. Finding the min-embedding helps to remove the redundant dimensions of the original embedding.
3. For a given function $f : \mathcal{M} \to \mathbb{R}$, find alpha $\alpha \in \mathbb{R}^k$ such that $f \approx \alpha \cdot \phi$. Just least square regression, linear transformer is capable of doing that [ZFB].
4. Predict $f(x)$ using $f \approx \alpha \cdot \phi$. This is basically an in-context learning step.

Relevant literature

- Geshkovski et al. [Ges+25] – models transformers as interacting particle systems
- Belkin and Niyogi [BN06] – on the convergence of discrete Laplacian to Laplace-Beltrami operator
- Zhang and Zhou [ZZ25] – PDE view on transformers
- Jacot, Gabriel, and Hongler [JGH20]

8 Transformers learn ReLU in context

Assumptions

- Bounded domain $\mathcal{M} \subset R^k$
- Function class $\mathcal{F} = \{f(\mathbf{x}) \mid f : \mathcal{M} \to \mathbb{R}, f(\mathbf{x}) = \sum_{i=1}^m a_i \mathrm{ReLU}(\mathbf{w}_i^\top \mathbf{x}) \text{ where } a_i \in \mathbb{R}, \mathbf{w}_i \in \mathbb{R}^k\}$
- Assume that $\dim(\mathcal{F}) = d \ll km$, specifically, each $\mathbf{w}_i \in \mathrm{span}\{\boldsymbol{\omega}_1, \ldots, \boldsymbol{\omega}_d\}$

Problem Set Up

Given $\{(\mathbf{x}_i, y_i = f(\mathbf{x}_i))\}$ we want to recover $\bar{a}, W$ that parametrize $f_{\bar{a},W}$. Let

$$Q = I - \sum_{i=1}^{d} \boldsymbol{\omega}_i \boldsymbol{\omega}_i^\top$$

$$\min_{\substack{u_j : \mathcal{M} \to \mathbb{R}^k \\ j=1,\ldots,m}} \int_{\mathbf{x} \in \mathcal{M}} \underbrace{\left\| \sum_{j=1}^{m} u_j(\mathbf{x})^\top \mathbf{x} - y(\mathbf{x}) \right\|_2^2}_{\text{function fitting error}} + \underbrace{\lambda \left\| \nabla u_j \right\|_Q^2 dx}_{\text{piecewise linearity}}$$

Convex in u_j hence, can be solved by gradient descent:

$$\frac{\partial u_j}{\partial t}(\mathbf{x}) = 2 \left(\sum_{i=1}^{m} u_i(\mathbf{x})^\top \mathbf{x} - y(\mathbf{x}) \right) \mathbf{x} - \lambda \Delta_Q u_j(\mathbf{x})$$

$$u_j^{(l+1)} = u_j^{(l)} - \eta \frac{\partial u_j^{(l)}}{\partial t}$$

where Δ_Q is the Laplacian with respect to the metric Q.

$$\min_{u_j} \int \left\| \sum_j u_j^\top \mathbf{x} - y \right\|^2 + \lambda \left\| \sum_j \nabla u_j \nabla u_j^\top \right\|_* dx \tag{1}$$

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9 Intro